8 (a) (i)

$$0 = 4mV + (A-4)mv \quad \text{B1}$$
$$4V = -(a-4)v \quad \text{B1}$$

(ii)

$$ratio = \frac{4mV^2}{4(A-4)mv^2} \qquad (1) \quad \text{B1}$$

From (i)

$$V = -\left(\frac{A-4}{4}\right)v \quad \text{B1}$$

Sub into (1)

$$ratio = \frac{4(\frac{A-4}{4})^2 v^2}{(A-4)v^2} \quad \text{B1}$$
$$ratio = \frac{A}{4} - 1$$

b (i)

$$E_{released} = [m_{Bi} - (m_\alpha + m_{Th})]c^2 \quad \text{C1}$$
$$= [211.9459 - (4.0015 + 207.9374)]1.66 \times 10^{-27} \frac{(3.00 \times 10^8)^2}{1.60 \times 10^{-23}} \quad \text{C1 C1}$$

= 654 MeV A1

(ii)

$$\frac{E_\propto}{E_{Th}} = \frac{A}{4} - 1$$
$$\frac{E_\propto}{6.54 - E_\propto} = \frac{212}{4} - 1 = 52 \quad \text{C1}$$
$$\frac{6.54}{E_\propto} - 1 = \frac{1}{52}$$
$$E_\propto = 6.42\ MeV \quad \text{A1}$$

c (i) Gamma ray is also emitted in the process. Hence energy of alpha particle is smaller B1

(ii) Gamma ray is likely to carry off part of the momentum in a direction different from that of thallium nucleus and alpha particle. By conservation of momentum, the 2 particles cannot move in opposite direction B1 B1 B1

(ii) The expression would apply to an isolated charge. But there are two charged objects here.

d	**(i)**	3600s	A1
	(ii)	Two hours is equivalent to two half lives of bismuth	B1
		hence number of radioactive bismuth nuclei remaining is approx. N/4,	B1
		forming 3N/4 thallium nuclei.	B1
		However, thallium has a much smaller half life and would decay quickly to form other nuclei hence number of thallium nuclei is less than 3N/4.	